



UNIVERSITI MALAYSIA TERENGGANU

FACULTY OF COMPUTER SCIENCE AND MATHEMATICS

COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH HONORS

SEMESTER II 2025/2026

CSA3023

WEB-BASED APPLICATION DEVELOPMENT

LAB 6:

JSP: SAVING AND RETRIEVING DATA FROM DATABASE

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Prepared for:

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Date:

17 MAY 2026

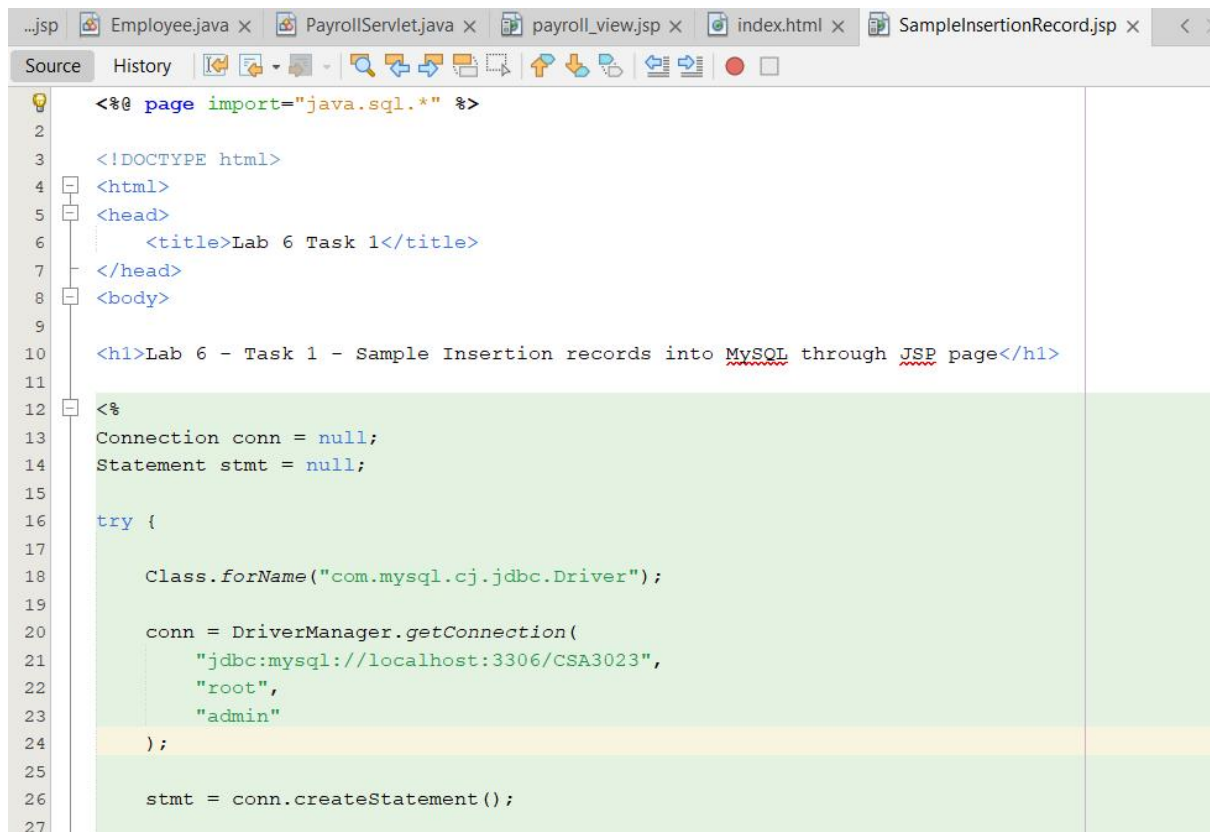
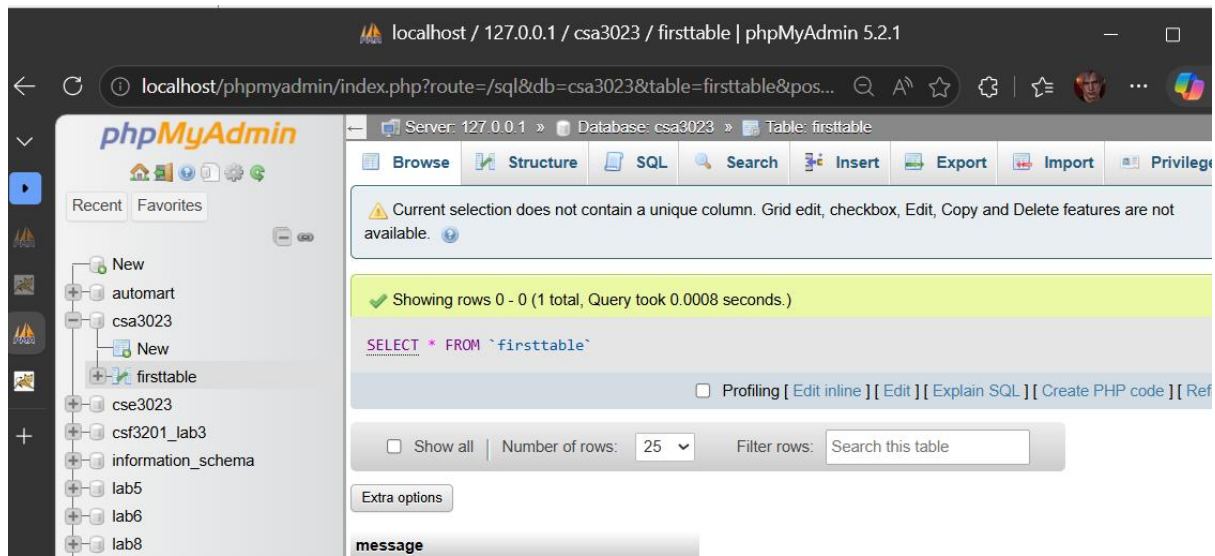
Task 1: Using JSP Page to Access a Simple MySQL Database

Objective

Write a JSP that can insert data to MYSQL database as “Welcome to access MySQL database with JSP...!” and also display steps of how to connect with MYSQL database.

Problem Description

- Create a table known as FirstTable using database schema CSA3203, create the first column as a character length 45.
- Create SampleInsertionRecord.jsp page to process and acknowledge the user upon inserting record in the database



```

27
28     String sql = "INSERT INTO FirstTable VALUES('Welcome to access MySQL database with JSP...!')";
29
30     stmt.executeUpdate(sql);
31
32     out.println("Record inserted successfully!");
33
34 } catch(Exception e) {
35
36     out.println(e);
37
38 } finally {
39
40     if(stmt != null)
41         stmt.close();
42
43     if(conn != null)
44         conn.close();
45 }
46 %>
47
48 </body>
49 </html>

```



Reflection

1. How do you want to submit specific information from one form to next form?
Answer:
Specific information can be submitted from one form to another using HTML form elements and retrieving the data using `request.getParameter()` in JSP.
2. What happened if the field name you specify in `request.getParameter("field_name")` in second page is different from the field name you defined in first page?
Answer:

If the field name is different, the system cannot retrieve the data correctly and the returned value will be null.

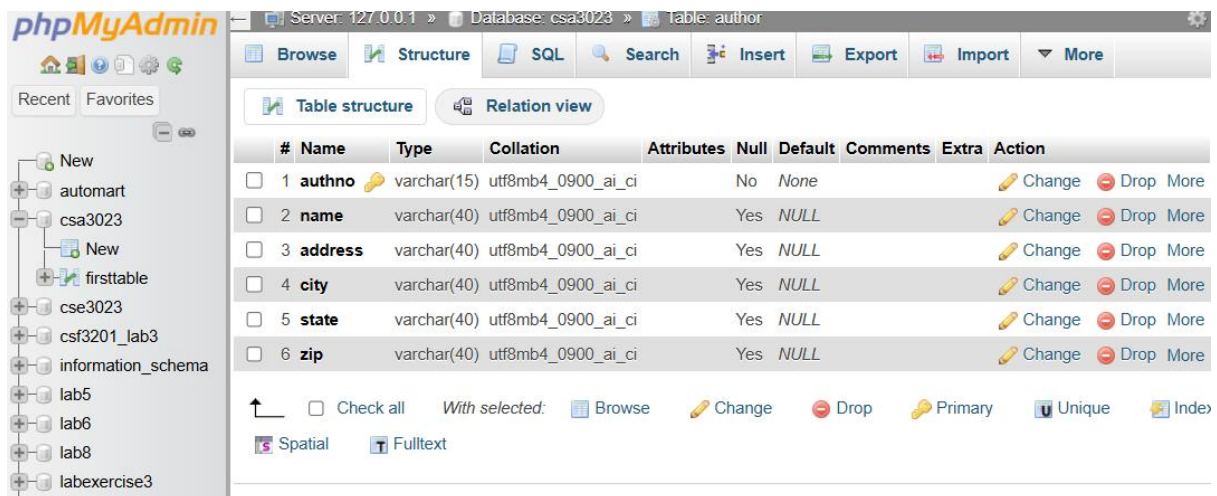
Task 2: Create Records via JSP Page

Objective

Using JSP to insert records retrieve from MySQL database.

Problem Description

1. Create a table known as Author using database schema CSA3023 using these attributes:
 - authno as a character length 15 and must be primary key name.
 - address as a character length 40.
 - city as a character length 40.
 - state as a character length 40.
 - zip as a character length 40.
2. Create insertAuthor.jsp as a main interface to register a new author.
3. Create processAuthor.jsp page to process and acknowledge the user upon inserting record in the database.



The screenshot shows the phpMyAdmin interface. On the left is a sidebar with a tree view of databases and tables. The main panel displays the 'Table structure' for the 'author' table in the 'csa3023' database. The table has six columns: #, Name, Type, Collation, Attributes, Null, Default, Comments, Extra, and Action. The columns are: 1. authno (varchar(15), utf8mb4_0900_ai_ci, No, None, Primary key), 2. name (varchar(40), utf8mb4_0900_ai_ci, Yes, NULL), 3. address (varchar(40), utf8mb4_0900_ai_ci, Yes, NULL), 4. city (varchar(40), utf8mb4_0900_ai_ci, Yes, NULL), 5. state (varchar(40), utf8mb4_0900_ai_ci, Yes, NULL), and 6. zip (varchar(40), utf8mb4_0900_ai_ci, Yes, NULL). At the bottom, there are checkboxes for 'Check all', 'With selected', and buttons for 'Browse', 'Change', 'Drop', 'Primary', 'Unique', 'Index', 'Spatial', and 'Fulltext'.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	authno	varchar(15)	utf8mb4_0900_ai_ci		No	None			Change Drop More
2	name	varchar(40)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
3	address	varchar(40)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
4	city	varchar(40)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
5	state	varchar(40)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
6	zip	varchar(40)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More

...jsp index.html x SampleInsertionRecord.jsp x insertAuthor.jsp x Author.java x processAuthor.jsp x

Source History

```
1  /*
2  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this li
3  * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
4  */
5  package lab6.com;
6  public class Author {
7
8      private String authno;
9      private String name;
10     private String address;
11     private String city;
12     private String state;
13     private String zip;
14
15     public String getAuthno() {
16         return authno;
17     }
18
19     public void setAuthno(String authno) {
20         this.authno = authno;
21     }
22
23     public String getName() {
24         return name;
25     }
26
27     public void setName(String name) {
28         this.name = name;
29
30
31     public String getAddress() {
32         return address;
33     }
34
35     public void setAddress(String address) {
36         this.address = address;
37     }
38
39     public String getCity() {
40         return city;
41     }
42
43     public void setCity(String city) {
44         this.city = city;
45     }
46
47     public String getState() {
48         return state;
49     }
50
51     public void setState(String state) {
52         this.state = state;
53     }
54
55     public String getZip() {
56         return zip;
57     }
58
59     public void setZip(String zip) {
60         this.zip = zip;
61     }
62 }
```

```
..jsp index.html x SampleInsertionRecord.jsp x insertAuthor.jsp x Author.java x processAuthor.jsp x
Source History
1 <!--
2     Document    : insertAuthor
3     Created on  : 17 May 2026, 4:47:10 pm
4     Author     : nursyaheera
5 -->
6
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <!DOCTYPE html>
9 <html>
10 <head>
11     <title>Insert Author</title>
12 </head>
13 <body>
14
15     <h1>Register New Author</h1>
16
17     <form action="processAuthor.jsp" method="post">
18
19         Author No:
20         <input type="text" name="authno"><br><br>
21
22         Name:
23         <input type="text" name="name"><br><br>
24
25         Address:
26         <input type="text" name="address"><br><br>
27
28         City:
29         <input type="text" name="city"><br><br>
30
31         State:
32         <input type="text" name="state"><br><br>
33
34         Zip:
35         <input type="text" name="zip"><br><br>
36
37         <input type="submit" value="Submit">
38         <input type="reset" value="Cancel">
39
40     </form>
41
42 </body>
43 </html>
```

```
...jsp index.html x SampleInsertionRecord.jsp x insertAuthor.jsp x Author.java x processAuthor.jsp x
Source History
1 <!--
2     Document    : processAuthor
3     Created on  : 17 May 2026, 4:49:06 pm
4     Author      : nursyaheera
5 -->
6
7 <%@ page import="java.sql.*" %>
8 <%@ page import="lab6.com.Author" %>
9
10 <jsp:useBean id="author" class="lab6.com.Author" />
11
12 <jsp:setProperty name="author" property="*" />
13
14 <!DOCTYPE html>
15 <html>
16 <head>
17     <title>Process Author</title>
18 </head>
19 <body>
20
21 <%
22
23 Connection conn = null;
24 PreparedStatement ps = null;
25
26 try {
27
28     Class.forName("com.mysql.cj.jdbc.Driver");
29
30     conn = DriverManager.getConnection(
31         "jdbc:mysql://localhost:3306/CSA3023",
32         "root",
33         "admin"
```

```

34     );
35
36     String sql = "INSERT INTO Author VALUES (?, ?, ?, ?, ?, ?)";
37
38     ps = conn.prepareStatement(sql);
39
40     ps.setString(1, author.getAuthno());
41     ps.setString(2, author.getName());
42     ps.setString(3, author.getAddress());
43     ps.setString(4, author.getCity());
44     ps.setString(5, author.getState());
45     ps.setString(6, author.getZip());
46
47     ps.executeUpdate();
48
49     out.println("Author inserted successfully!");
50
51 } catch(Exception e) {
52
53     out.println(e);
54
55 } finally {
56
57     if(ps != null)
58         ps.close();
59
60     if(conn != null)
61         conn.close();
62 }
63 %>
64
65 </body>
66 </html>

```

 Insert Author



localhost:8081/Lab6/insertAuthor.jsp







Register New Author

Author No:
 Name:
 Address:
 City:
 State:
 Zip:

Insert Author

localhost:8081/Lab6/insertAuthor.jsp Summarize

Register New Author

Author No:

Name:

Address:

City:

State:

Zip:

Process Author

localhost:8081/Lab6/processAuthor.jsp

Author inserted successfully!

Reflection

1. What have you learnt from this exercise?

Answer:

Learned how to connect JSP with MySQL database and insert records using PreparedStatement and JavaBeans.

2. Define step by step before you successfully perform the transaction in a database.

Answer:

- Start Apache and MySQL
- Create database
- Create table
- Add MySQL Connector
- Create JSP form
- Retrieve form data
- Connect database
- Execute SQL query
- Close connection

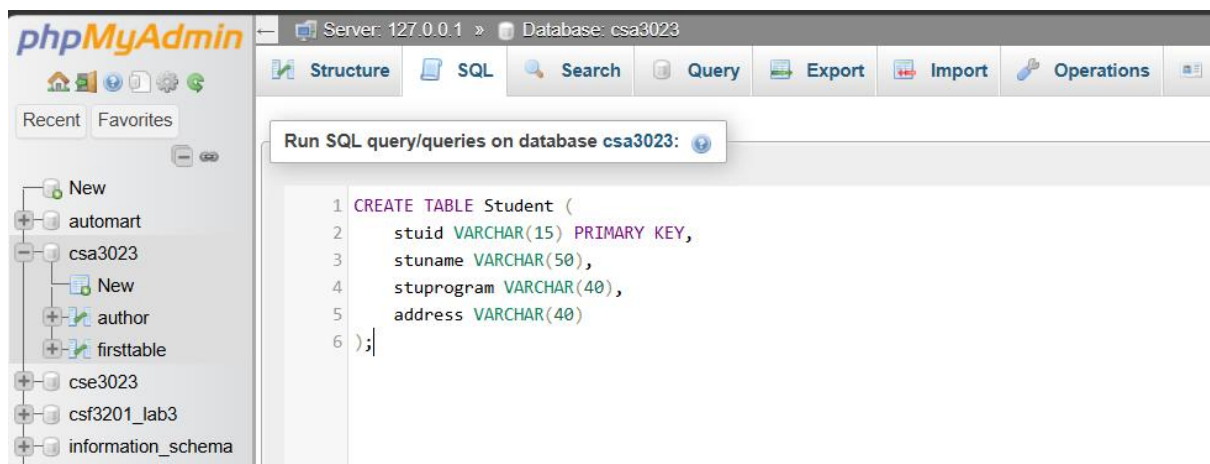
Task 3: Create Records Constrained by Regular Expression in JSP

Objective

Using JSP Standard Action, scriptlets and regular expression to insert records retrieve from MySQL database.

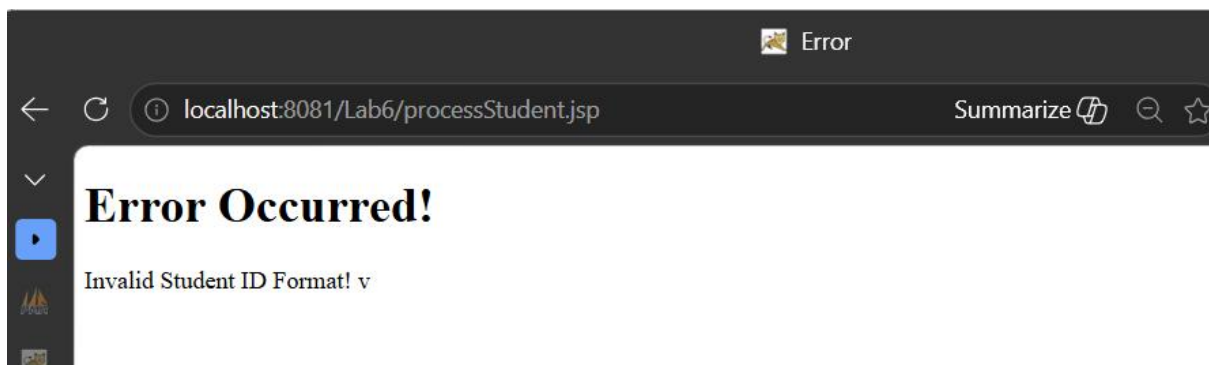
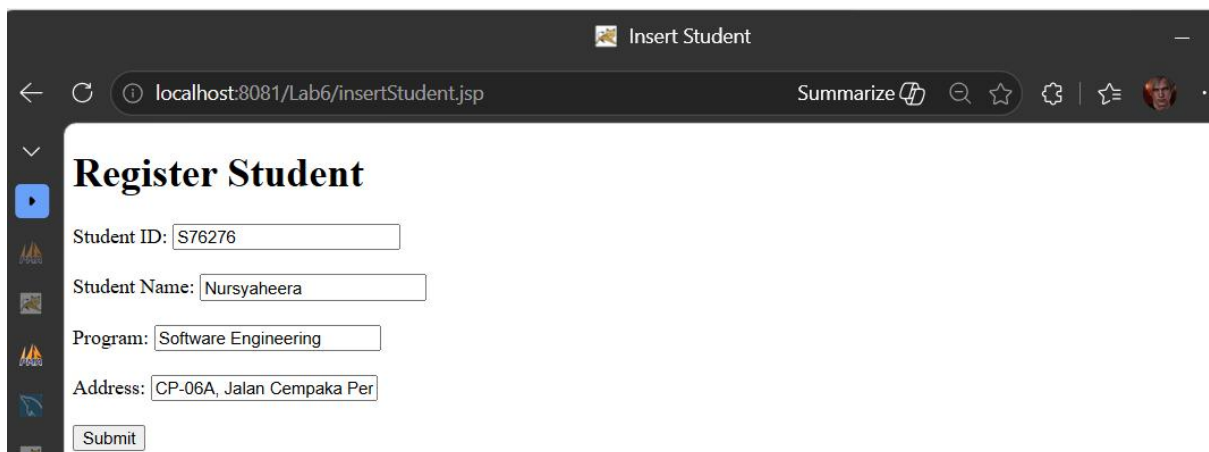
Problem Description

1. Create a table known as Student using database schema CSA3023 using these attributes:
 - stuid as a character length 15 and must be primary key name.
 - stuname as a character length 50.
 - stuprogram as a character length 40.
 - address as a character length 40.
2. Create insertStudent.jsp as a main interface to register new book.
3. Create processStudent.jsp page to process and acknowledge the user upon inserting record in the database.
4. Create displayStudent.jsp page to populate records.
5. Create errorStudent.jsp page to handle an error.



```
...va processAuthor.jsp x Student.java x insertStudent.jsp x processStudent.jsp x errorStudent.jsp x
Source History
1 <%--
2 Document : processStudent
3 Created on : 17 May 2026, 5:26:01 pm
4 Author : nursyaheera
5 --%>
6
7 <%@ page import="java.sql.*" %>
8 <%@ page errorPage="errorStudent.jsp" %>
9 <%@ page import="lab6.com.Student" %>
10
11 <jsp:useBean id="student" class="lab6.com.Student" />
12
13 <%
14 student.setStuid(request.getParameter("stuid"));
15 student.setStuname(request.getParameter("stuname"));
16 student.setStuprogram(request.getParameter("stuprogram"));
17 student.setAddress(request.getParameter("address"));
18
19 Connection conn = null;
20 PreparedStatement ps = null;
21
22 Class.forName("com.mysql.cj.jdbc.Driver");
23
24 conn = DriverManager.getConnection(
25     "jdbc:mysql://localhost:3306/CSA3023",
26     "root",
27     ""
28 );
29
30 String sql = "INSERT INTO Student VALUES (?, ?, ?, ?)";
31
32 ps = conn.prepareStatement(sql);
33
34 ps.setString(1, student.getStuid());
35 ps.setString(2, student.getStuname());
36 ps.setString(3, student.getStuprogram());
37 ps.setString(4, student.getAddress());
38
39 ps.executeUpdate();
40
41 out.println("Student registered successfully!");
42
43 ps.close();
44 conn.close();
45 %>
```

```
...va processAuthor.jsp x Student.java x insertStudent.jsp x processStudent.jsp x errorStudent.jsp x
Source History
1 <!--
2     Document    : errorStudent
3     Created on  : 17 May 2026, 5:26:32 pm
4     Author     : User
5 -->
6
7 <%@ page isErrorPage="true" %>
8
9 <!DOCTYPE html>
10 <html>
11 <head>
12     <title>Error</title>
13 </head>
14 <body>
15
16     <h1>Error Occurred!</h1>
17
18     <%= exception.getMessage() %>
19
20 </body>
21 </html>v
```



Reflection

1. What have you learnt from this exercise?

Answer:

Learned how to use JSP Standard Action, JavaBeans, regular expressions, and JDBC to validate user input and insert records into a MySQL database.

2. Define step by step before you successfully perform the transaction in a database

Answer:

1. Create the database and table in phpMyAdmin.
2. Create the JSP form interface.
3. Create JavaBeans class with getter and setter methods.
4. Apply regular expression validation.
5. Retrieve form data using JSP.
6. Load JDBC driver.
7. Create database connection.
8. Prepare SQL statement using PreparedStatement.
9. Execute SQL query.
10. Close database connection.

Task 4: Perform Retrieving Records Via JSP Page

Objective

Use Java Scriptlet to query a list of records.

Problem Description

Retrieve student records and populate in the table.

```
...jsp x Student.java x insertStudent.jsp x processStudent.jsp x errorStudent.jsp x queryStudent.jsp x
Source History
1 <!--
2     Document    : queryStudent
3     Created on  : 17 May 2026, 5:33:34 pm
4     Author      : nursyaheera
5 -->
6
7 <%@ page import="java.sql.*" %>
8
9 <!DOCTYPE html>
10 <html>
11 <head>
12     <title>Query Student</title>
13
14     <style>
15
16         table {
17             border-collapse: collapse;
18             width: 80%;
19         }
20
21         th, td {
22             border: 1px solid black;
23             padding: 10px;
24         }
25
26     </style>
27
28 </head>
29 <body>
30
31 <h1>Lab 6 - Task 4: Retrieving record via JSP Page</h1>
32
33 <table>
```

```

34
35 <tr>
36     <th>Student ID</th>
37     <th>Name</th>
38     <th>Program</th>
39     <th>Address</th>
40 </tr>
41
42 <%
43
44 Connection conn = null;
45 Statement stmt = null;
46 ResultSet rs = null;
47
48 try {
49
50     Class.forName("com.mysql.cj.jdbc.Driver");
51
52     conn = DriverManager.getConnection(
53         "jdbc:mysql://localhost:3306/CSA3023",
54         "root",
55         "admin"
56     );
57
58     stmt = conn.createStatement();
59
60     rs = stmt.executeQuery("SELECT * FROM Student");
61
62     while(rs.next()) {
63
64 %>
65
66 <tr>
67     <td><%= rs.getString("stuid") %></td>
68     <td><%= rs.getString("stuname") %></td>
69     <td><%= rs.getString("stuprogram") %></td>
70     <td><%= rs.getString("address") %></td>
71 </tr>
72
73 <%
74     }
75
76 } catch(Exception e) {
77
78     out.println(e);
79
80 }
81 %>
82
83 </table>
84
85 </body>
86 </html>

```

phpMyAdmin

Server: 127.0.0.1 » Database: csa3023 » Table: student

Browse Structure SQL Search Insert Export Import Privileges

Show query box

✓ 1 row inserted. (Query took 0.0285 seconds.)

```
INSERT INTO Student VALUES ('UK12489', 'Ahmad Salam', 'BSc with IM', 'Terengganu');
```

[Edit inline] [Edit] [Create F

✓ 1 row inserted. (Query took 0.0044 seconds.)

```
INSERT INTO Student VALUES ('UK56789', 'Rosnah Azman', 'BSc Soft. Eng.', 'Kelantan');
```

[Edit inline] [Edit] [Create F

✓ 1 row inserted. (Query took 0.0048 seconds.)

```
INSERT INTO Student VALUES ('UK67342', 'Liew Cheng Huat', 'BSc in Robotics', 'Pahang');
```

[Edit inline] [Edit] [Create F

Query Student

localhost:8081/Lab6/queryStudent.jsp Summarize

Lab 6 - Task 4: Retrieving record via JSP Page

Student ID	Name	Program	Address
UK12489	Ahmad Salam	BSc with IM	Terengganu
UK56789	Rosnah Azman	BSc Soft. Eng.	Kelantan
UK67342	Liew Cheng Huat	BSc in Robotics	Pahang

Reflection

1. What have you learnt from this exercise?

Answer:

Learned how to retrieve records from MySQL database using JSP scriptlet and display the records in an HTML table.

2. Describe the benefits of using JavaBeans.

Statement	PreparedStatement
Less secure	More secure
Vulnerable SQL Injection	Prevent SQL Injection

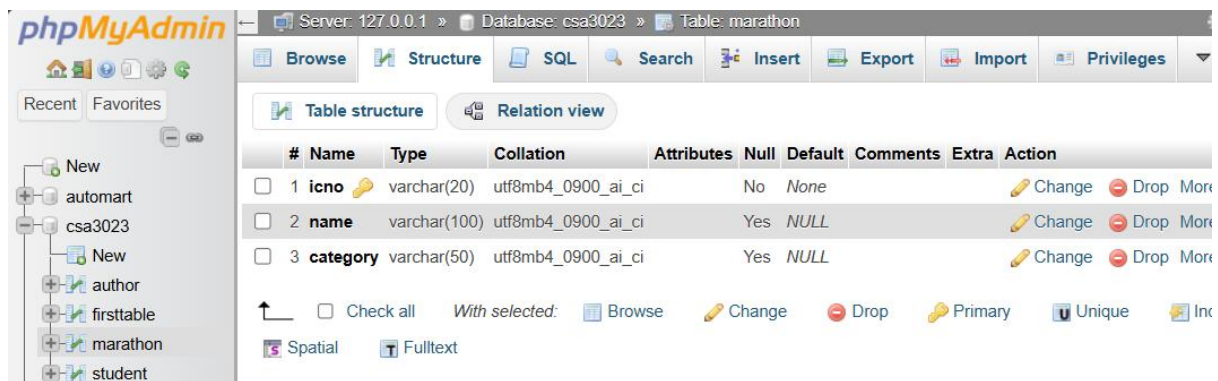
Task 5: Create A Record Using JSP Model 1

Objective

Use JavaBeans to perform SQL transaction.

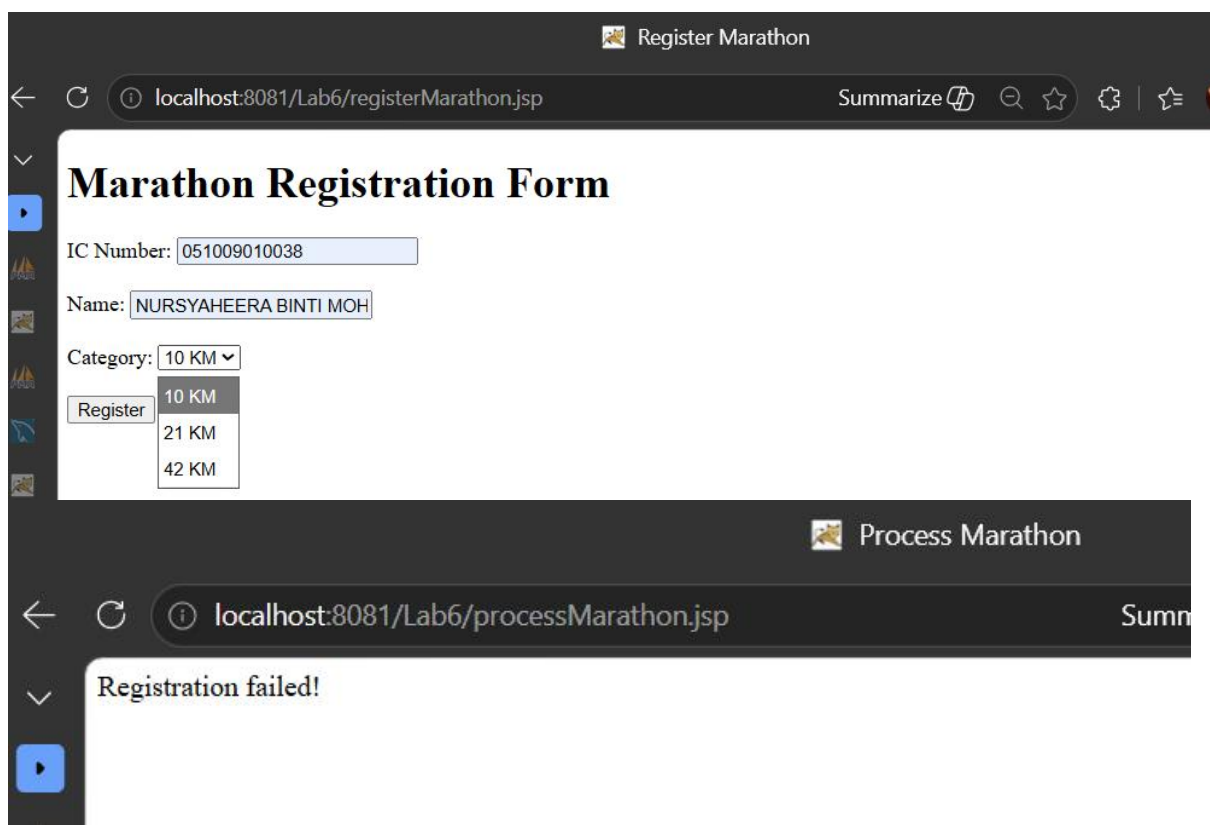
Problem Description

Create a sample web form to register the Marathon event.



The screenshot shows the phpMyAdmin interface for a database named 'csa3023'. The 'marathon' table is selected, and its structure is displayed in 'Table structure' view. The table has three columns: 'icno' (varchar(20)), 'name' (varchar(100)), and 'category' (varchar(50)). All columns use the 'utf8mb4_0900_ai_ci' collation. The 'icno' column is not null and has a default value of 'None'. The 'name' and 'category' columns are nullable with a default value of 'NULL'. The interface includes a sidebar with a tree view of the database structure, showing 'csa3023' with sub-items like 'author', 'firsttable', 'marathon', and 'student'. The top navigation bar includes links for 'Browse', 'Structure', 'SQL', 'Search', 'Insert', 'Export', 'Import', and 'Privileges'.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	icno	varchar(20)	utf8mb4_0900_ai_ci		No	None			Change Drop More
2	name	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
3	category	varchar(50)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More



The screenshot shows two web browser windows. The top window is titled 'Register Marathon' and displays a form for registering a marathon. The form has three input fields: 'IC Number' with the value '051009010038', 'Name' with the value 'NURSYAHEERA BINTI MOH', and 'Category' with a dropdown menu showing '10 KM'. Below the 'Category' dropdown is a 'Register' button. The bottom window is titled 'Process Marathon' and displays a message 'Registration failed!'.

Register Marathon

IC Number:

Name:

Category:

Process Marathon

Registration failed!

Reflection

1. What have you learnt from this exercise?
Answer: Learned how to use JSP Model 1 architecture with JavaBeans and DAO to perform database transactions.
2. Describe the benefits of using JavaBeans.

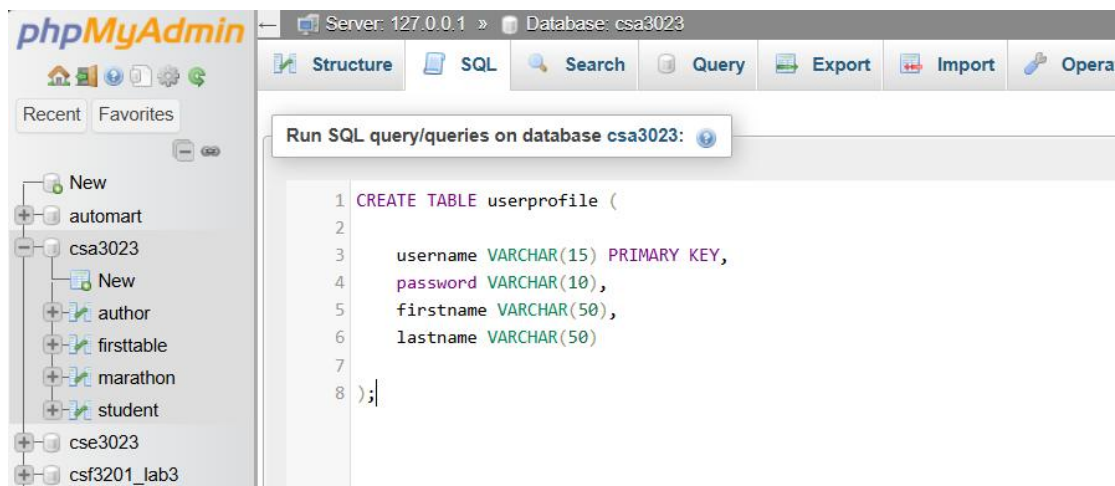
Answer:

- Reusable code
- Easier maintenance
- Better data encapsulation
- Cleaner project structure
- Separation between business logic and presentation layer

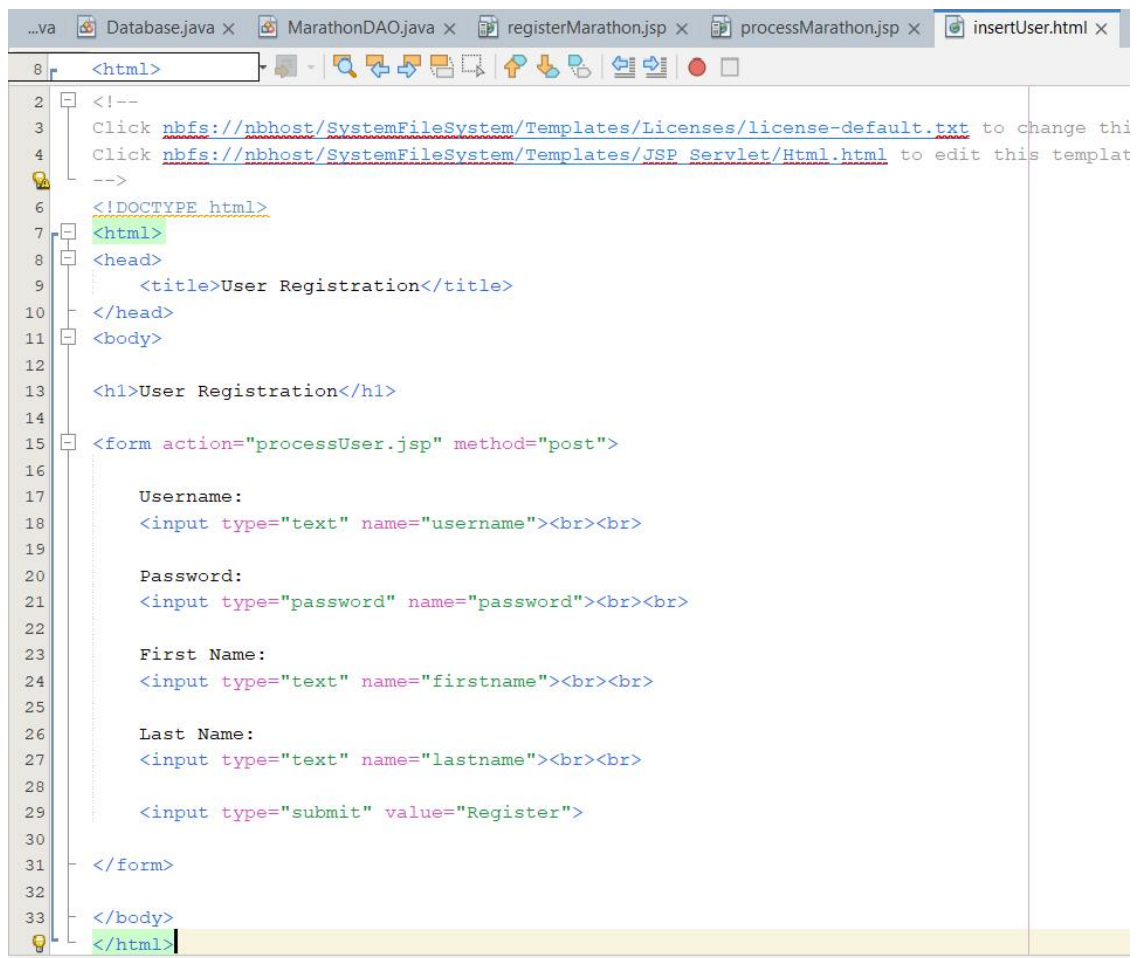
Exercise

Implement user login

1. Create a table known as userprofile using database schema CSA3023 using these attributes
 - username as a character length 15 and must be primary key
 - password as a character length 10
 - firstname as varchar(50)
 - lastname as varchar(50)



2. Create insertUser.html as the main interface to register a new user.

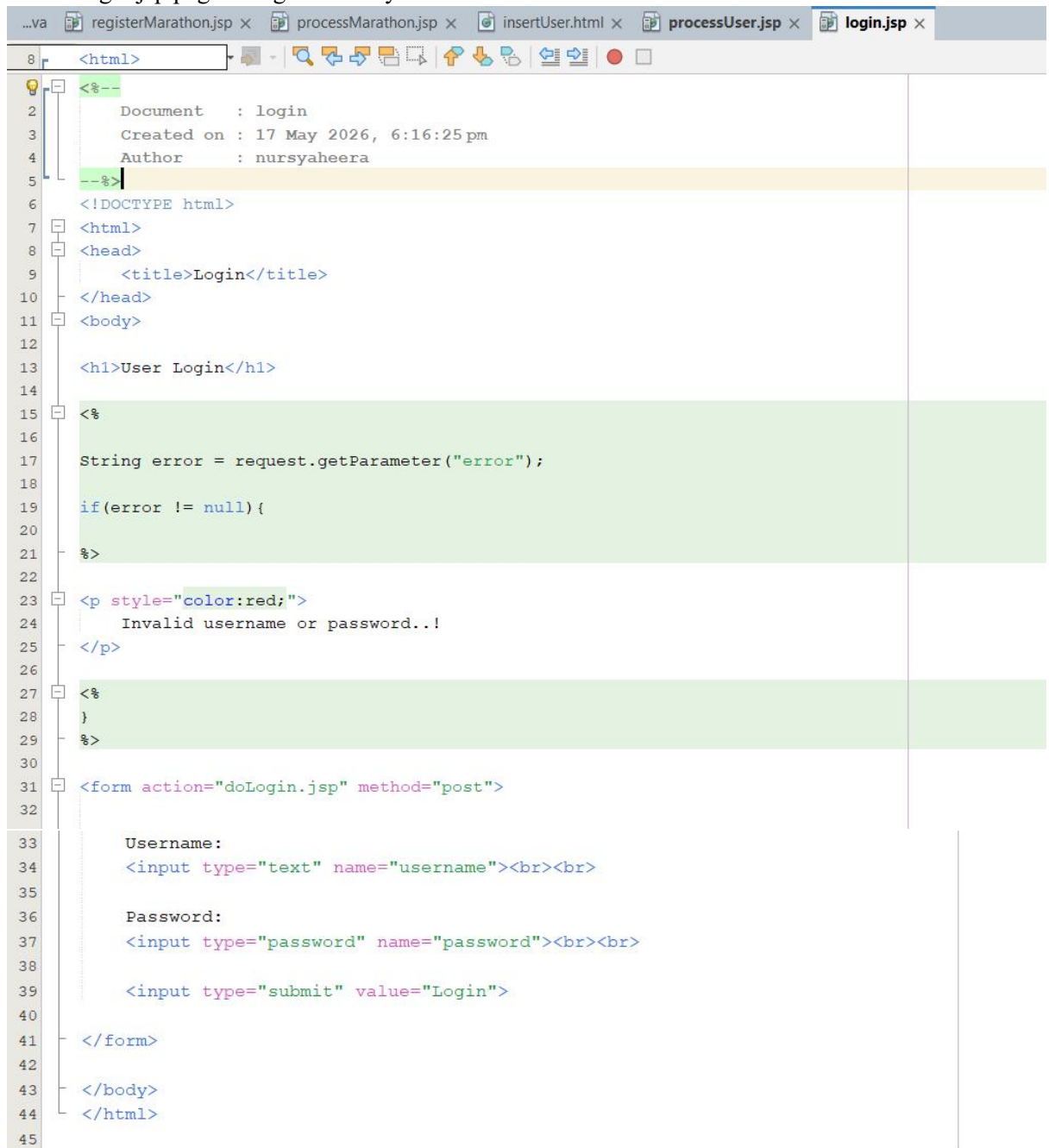


```
...va Database.java x MarathonDAO.java x registerMarathon.jsp x processMarathon.jsp x insertUser.html x
8 <html>
2 <!--
3 Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change thi
4 Click nbfs://nbhost/SystemFileSystem/Templates/JSP\_Servlet/Html.html to edit this templat
5 -->
6 <!DOCTYPE html>
7 <html>
8 <head>
9     <title>User Registration</title>
10 </head>
11 <body>
12
13 <h1>User Registration</h1>
14
15 <form action="processUser.jsp" method="post">
16
17     Username:
18     <input type="text" name="username"><br><br>
19
20     Password:
21     <input type="password" name="password"><br><br>
22
23     First Name:
24     <input type="text" name="firstname"><br><br>
25
26     Last Name:
27     <input type="text" name="lastname"><br><br>
28
29     <input type="submit" value="Register">
30
31 </form>
32
33 </body>
34 </html>
```

3. Create processUser.jsp page to process the record.

```
...va  MarathonDAO.java x  registerMarathon.jsp x  processMarathon.jsp x  insertUser.html x  processUser.jsp x
8      <html>
9      <!--
10         Document    : processUser
11         Created on   : 17 May 2026, 6:15:16 pm
12         Author      : nursyaera
13     -->
14
15     <%@ page import="java.sql.*" %>
16
17     <!DOCTYPE html>
18     <html>
19     <head>
20         <title>Process User</title>
21     </head>
22     <body>
23
24     <%
25
26     Connection conn = null;
27     PreparedStatement ps = null;
28
29     try{
30
31         String username = request.getParameter("username");
32         String password = request.getParameter("password");
33         String firstname = request.getParameter("firstname");
34         String lastname = request.getParameter("lastname");
35
36         Class.forName("com.mysql.cj.jdbc.Driver");
37
38         conn = DriverManager.getConnection(
39             "jdbc:mysql://localhost:3306/CSA3023",
40             "root",
41             "admin"
42         );
43
44         String sql =
45             "INSERT INTO userprofile VALUES (?, ?, ?, ?)";
46
47         ps = conn.prepareStatement(sql);
48
49         ps.setString(1, username);
50         ps.setString(2, password);
51         ps.setString(3, firstname);
52         ps.setString(4, lastname);
53
54         ps.executeUpdate();
55
56         out.println("User registered successfully!");
57     }catch(Exception e){
58         out.println(e);
59     }finally{
60         if(ps != null)
61             ps.close();
62
63         if(conn != null)
64             conn.close();
65     }
66     %>
67 </body>
68 </html>
```

4. Create login.jsp page to login to the system.



```
1 <!--
2     Document    : login
3     Created on  : 17 May 2026, 6:16:25 pm
4     Author      : nursyaheera
5 -->
6 <!DOCTYPE html>
7 <html>
8 <head>
9     <title>Login</title>
10 </head>
11 <body>
12
13     <h1>User Login</h1>
14
15     <%
16
17     String error = request.getParameter("error");
18
19     if(error != null){
20
21     %>
22
23     <p style="color:red;">
24         Invalid username or password..!
25     </p>
26
27     <%
28     }
29     %>
30
31     <form action="doLogin.jsp" method="post">
32
33         Username:
34         <input type="text" name="username"><br><br>
35
36         Password:
37         <input type="password" name="password"><br><br>
38
39         <input type="submit" value="Login">
40
41     </form>
42
43 </body>
44 </html>
45
```

5. Create doLogin.jsp to validate username and password. If validation succeeds, redirect the page to main.jsp page that displays the username, firstname, and lastname

```
...jsp x processMarathon.jsp x insertUser.html x processUser.jsp x login.jsp x doLogin.jsp x main.jsp x
8 <html>
9 <!--
10 Document : doLogin
11 Created on : 17 May 2026, 6:17:29 pm
12 Author : nursyaheera
13 -->
14
15 <%@ page import="java.sql.*" %>
16
17 <%
18
19 Connection conn = null;
20 PreparedStatement ps = null;
21 ResultSet rs = null;
22
23 try{
24
25     String username = request.getParameter("username");
26     String password = request.getParameter("password");
27
28     Class.forName("com.mysql.cj.jdbc.Driver");
29
30     conn = DriverManager.getConnection(
31         "jdbc:mysql://localhost:3306/CSA3023",
32         "root",
33         "admin"
34     );
35
36     String sql =
37         "SELECT * FROM userprofile WHERE username=? AND password=?";
38
39     ps = conn.prepareStatement(sql);
40
41     ps.setString(1, username);
42
43     ps.setString(2, password);
44
45     rs = ps.executeQuery();
46
47     if(rs.next()){
48
49         session.setAttribute("username",
50             rs.getString("username"));
51
52         session.setAttribute("firstname",
53             rs.getString("firstname"));
54
55         session.setAttribute("lastname",
56             rs.getString("lastname"));
57
58         response.sendRedirect("main.jsp");
59
60     }else{
61
62         response.sendRedirect("login.jsp?error=1");
63
64     }
65 }catch(Exception e){
66
67     out.println(e);
68
69 }
70 %>
```

```
...jsp x processMarathon.jsp x insertUser.html x processUser.jsp x login.jsp x doLogin.jsp x main.jsp x
8 | <html>
  | <!--
  |   Document    : main
  |   Created on  : 17 May 2026, 6:18:02 pm
  |   Author     : nursyaheera
  | -->
  |
  | <!DOCTYPE html>
  | <html>
  | <head>
  |   <title>Main Page</title>
  | </head>
  | <body>
  |   <h1>Welcome to the System</h1>
  |
  |   <p>
  |     Username:
  |     <%= session.getAttribute("username") %>
  |   </p>
  |
  |   <p>
  |     First Name:
  |     <%= session.getAttribute("firstname") %>
  |   </p>
  |
  |   <p>
  |     Last Name:
  |     <%= session.getAttribute("lastname") %>
  |   </p>
  |
  | </body>
  | </html>
```

6. However if fails, redirect the page to login.jsp with message 'Invalid username or password..!'

User Registration

localhost:8081/Lab6/insertUser.html

Summarize

User Registration

Username: Ahmad

Password:

First Name: ahmad

Last Name: ali

Register

← ↻ ⓘ localhost:8081/Lab6/processUser.jsp

✓ User registered successfully!

